

# JAK / TYK2: 8 of 33 completed trials missed their primary endpoint

8 completed trials across 4 sponsors and 5 molecules ran to the end and missed their primary endpoint on the sponsor's own posted result; 20 met it. Only 4 of 79 closed trials were stopped early for efficacy or safety, which is why this class is read through its completed trials rather than a stop rate. Most completed trials post nothing readable, so the count is a floor.

|                                                                       |                                        |                                                          |                                                         |
|-----------------------------------------------------------------------|----------------------------------------|----------------------------------------------------------|---------------------------------------------------------|
| 8 of 33                                                               | 4 / 5                                  | 5.1%                                                     | 62.2%                                                   |
| completed trials with a readable primary result missed it · 20 met it | sponsors / molecules behind the misses | stopped early: 4 of 79 closed trials (95% CI 2.0%–12.3%) | of 127 trials have closed · 48 still open or unresolved |

8 of 33

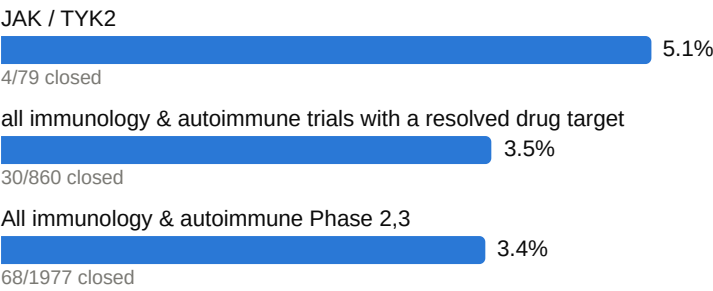
Completed and missed the primary endpoint. Of 65 completed trials in this segment, 33 posted a primary result on ClinicalTrials.gov that can be read — the sponsor's own comparison held to the sponsor's own threshold, or its statement that the endpoint was missed: 8 missed and 20 met, 5 split across co-primary endpoints. Counted separately and never in the rate above; a floor, not a rate. 9 of the 60 completed trials that finished more than a year ago have posted no results at all — a gap in what can be known, not a sign of failure.

## The completed trials that missed

| Trial       | Ph | Sponsor              | Experimental drugs      | Started | What the posted result says                                                                                                                                                                                                                                                                            |
|-------------|----|----------------------|-------------------------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NCT03934216 | 2  | Bristol Myers Squibb | CRAVACITINIB            | 2019-07 | Clinical Remission Response Rate at Week 12 — p 0.5935, no threshold stated; 0.2 or above, so not significant under any conventional threshold. Odds Ratio (OR) 0.9 (95% CI 0.3 to 2.4). Method: Stratified Cochran-Mantel-Haenszel.                                                                   |
| NCT03845517 | 2  | Pfizer               | BREPOCITINIB            | 2019-04 | Percentage of Participants Achieving SLE Responder Index (SRI) Change of 4 (SRI-4) at Week 52 — one-sided p 0.8076, no threshold stated; 0.2 or above, so not significant at any one-sided threshold in use. Risk Difference (RD) -7.5 (95% CI -24.5 to 9.5). Method: Cochran-Mantel-Haenszel.         |
| NCT03850483 | 2  | Pfizer               | BREPOCITINIB            | 2019-04 | Change From Baseline in Psoriasis Area and Severity Index (PASI) Score at Week 12 — p 0.1641 against the sponsor's stated threshold of 0.05: not significant. Least square (LS) mean difference -0.6 (90% CI -1.71 to 0.43). Method: ANCOVA.                                                           |
| NCT03616964 | 3  | Eli Lilly            | BARICITINIB             | 2018-08 | Percentage of Participants Achieving a Systemic Lupus Erythematosus Responder Index 4 (SRI-4) Response (4 mg Baricitinib) — p 0.711, no threshold stated; 0.1 or above, so not significant under any conventional threshold. Odds Ratio (OR) 1.07 (95% CI 0.75 to 1.53). Method: Regression, Logistic. |
| NCT03134222 | 2  | Gilead Sciences      | LANRAPLENIB, FILGOTINIB | 2017-05 | Change in Cutaneous Lupus Erythematosus Disease Area and Severity Index (CLASI) Activity Score From Baseline to Week 12 — p 0.75, no threshold stated; 0.2 or above, so not significant under any conventional threshold. Method: Mixed-effect repeated measures model.                                |
| NCT03046056 | 2  | Gilead Sciences      | FILGOTINIB              | 2017-04 | Percentage of Participants Who Achieved Clinical Remission at Week 24 — Risk Difference in Proportions 8.3 (90% CI -16.5 to 32.1), no p-value posted and no threshold stated: the interval is consistent with no effect under any conventional threshold, so not significant.                          |

The 6 most recently started of 8 completed trials that missed. The diligence report lists every one, with the numbers or the sentence each verdict was read from.

How this compares



**What is in this class:** 37 distinct experimental drugs across 127 trials (small molecule 127, unknown 3, antibody 2, protein 1). Most tested: FILGOTINIB (25), BARICITINIB (20), TOFACITINIB (19), UPADACITINIB (19), CRAVACITINIB (18). A class groups drugs by what they act on; check that the grouping is one you would make before reading the rate as a property of the mechanism.

**How much does this rest on one decision?** The 4 stops came from 3 sponsor-asset programmes across 3 sponsors; the largest (Eli Lilly) contributed 2; removing that programme's trials from both sides leaves 2.7%. Screened alongside 28 other segments, the rate is within what screening that many classes could produce (q=1); the molecules and trials that failed here are the record either way. A further 5 closed trials here were terminated with no cause recorded in the registry; if every one of them were biological the rate would be 11.4%. Read it as a screen worth checking against the underlying trials.

Bars show the share of closed trials stopped for biological reasons. Confidence intervals overlap where sample sizes are small — read the counts, not just the bars.

The stopped trials

| Trial       | Ph | Sponsor                               | Experimental drugs         | Stopped | Type         | Registry stop reason                                                                                                                                                                                                            |
|-------------|----|---------------------------------------|----------------------------|---------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NCT03599622 | 2  | Bristol Myers Squibb <span>BMJ</span> | CRAVACITINIB               | 2022-12 | Efficacy     | Trial terminated because of lack of efficacy in the short term acute phase                                                                                                                                                      |
| NCT03843125 | 3  | Eli Lilly <span>LLY</span>            | BARICITINIB                | 2022-04 | Benefit-risk | Study terminated due to insufficient evidence to support a positive benefit: risk profile. Safety findings were consistent with previously published OLUMIANT data                                                              |
| NCT03616912 | 3  | Eli Lilly <span>LLY</span>            | BARICITINIB                | 2021-11 | Benefit-risk | Study terminated due to insufficient evidence to support a positive benefit: risk profile in systemic lupus erythematosus patients.                                                                                             |
| NCT03823378 | 2  | AbbVie <span>ABBV</span>              | ELSUBRUTINIB, UPADACITINIB | 2020-09 | Efficacy     | Study M16-763 was terminated early as the benefit of each treatment arm from the feeder study (Study M16-063) did not provide appreciable evidence of differentiated clinical effect to warrant further long-term continuation. |

Method

Denominator: ClinicalTrials.gov interventional Phase 2,3 immunology & autoimmune trials started 2015–2024 that have closed (completed or terminated). Numerator: terminated trials whose registry stop reason is classified as biological (efficacy, safety or benefit-risk) by Classification V2 — held-out precision 95.5%, recall 95.3% (n=600). Drugs are linked to ChEMBL and the NCI Thesaurus; 48% of industry immunology & autoimmune trials in this window carry a resolved drug target, and a trial without one cannot enter a mechanism class. Intervals are Wilson 95%.

By start cohort

| Trials started | Stops | Closed | Rate  | 95% CI     |
|----------------|-------|--------|-------|------------|
| 2015–2017      | 0     | 36     | 0.0%  | 0.0%–9.6%  |
| 2018–2020      | 4     | 35     | 11.4% | 4.5%–26.0% |
| 2021–2024      | 0     | 8      | 0.0%  | 0.0%–32.4% |

Where the stops sit

Efficacy / futility: 2, Benefit-risk: 2.  
Sponsors with most stops: Eli Lilly (2), Bristol Myers Squibb (1), AbbVie (1).

Limits

Not a failure rate: completed trials that missed their endpoints are not in the rate. They are counted separately where the sponsor posted a primary result that can be read — most did not, so that count is a floor — and programmes discontinued after a completed trial do not appear. Stop reasons are sponsor-reported. This is a closed-trial proportion, not a time-to-event analysis: only 62.2% of trials in this segment have closed, and a trial that stops early enters the denominator sooner than one that runs to completion, which can inflate the rate in immature segments. Recent cohorts have fewer closed trials, so their rates are less stable. Research signals, not clinical or investment advice.

**The whole cohort, trial by trial.** The diligence report names every trial behind these figures — the ones still running and the ones that completed and missed their primary endpoint included — and places your molecule against each one that failed. €99, delivered immediately: [clinicaltrialfailures.com/asset-check](https://clinicaltrialfailures.com/asset-check)

